atmosphere. Only the comet theory proposed by Urey[13] approaches an adequate explanation.

I am indebted to Miss Maureen Sachs for the analytical determinations. The Director, South Australian Museum, Adelaide, provided sample *A*, and the Director, National Museum of Victoria, Melbourne, sent samples *B* and *C*. Without their generous co-operation, this investigation would not have been possible.

[1] Fenner, C., *Trans. Roy. Soc. South Australia*, **59**, 125 (1935).
[2] Fenner, C., *Trans. Roy. Soc. South Australia*, **58**, 62 (1934).
[3] Fenner, C., *Trans. Roy. Soc. South Australia*, **62**, 192 (1938).
[4] Baker, G., *Mem. Nat. Mus. Vict.*, **14**, 7 (1944).
[5] Baker, G., *Mem. Nat. Mus. Vict.*, **23**, 1 (1959).
[6] Hawkins, G. S., *Nature*, **185**, 300 (1960).
[7] Baker, G., and Gaskin, A. J., *J. Geol.*, **54**, 88 (1946).
[8] Ahrens, L. H., and Taylor, S. R., *Spectrochemical Analysis* (Reading, Mass.: Addison-Wesley 2nd Ed., 1961).
[9] Taylor, S. R., *Geochim. et Cosmochim. Acta* (in the press).
[10] Stevens, R. E., Niles, W. W., Chodos, A. A., Filby, R. H., Leininger, R. K., Ahrens, L. H., Fleischer, M., and Flanagan, F. J., *U.S. Geol. Survey Bull.*, **1113** (1960).
[11] Baker, G., *Mem. Nat. Mus. Vict.*, **20**, 59 (1956).
[12] Barnes, V. E., *Univ. Texas Publications*, **3945**, 477 (1940).
[13] Urey, H. C., *Nature*, **179**, 556 (1957).
[14] Cherry, R. D., and Taylor, S. R., *Geochim. et Cosmochim. Acta* (in the press).
[15] Cherry, R. D., Taylor, S. R., and Sachs, M., *Nature*, **187**, 680 (1960).
[16] Taylor, S. R., and Sachs, M., *Nature*, **188**, 387 (1960).
[17] Cohen, A. J., *Report Twenty-first Session, Inter. Geol. Congr. (Norden)*, **1**, 30 (1960).

PALÆOMAGNETISM IN SOME EAST GREENLAND SEDIMENTARY ROCKS

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DURING a visit to Central East Greenland in 1957, on the invitation of Dr. Lauge Koch[1], we collected specimens of sedimentary rocks, with the view of examining their magnetism. This article summarizes the results from the late Pre-Cambrian (two horizons), from the Permian, and from the Middle and Upper Trias. Other rocks collected (igneous rocks, lowest late-Pre-Cambrian horizon, Devonian, Lower Carboniferous, etc.) require further investigation, some being unstable. All the results, with a full discussion, will be published in *Meddelelser om Grønland*, with stratigraphical details and references.

Specimens were oriented in the field by rock compass[2]. In the laboratory the rocks were re-oriented, cored and measured on the astatic magnetometer housed at the Department of Geodesy and Geophysics, Cambridge. For these facilities we are indebted to Mr. B. C. Browne and Dr. J. C. Belshé. Afterwards, a second magnetometer was constructed for this work in the Sedgwick Museum, Cambridge. From each rock one or more samples were measured, and when necessary the vectors combined by taking means of the direction cosines to give a single mean vector for each rock, this mean vector forming the basic unit. After correcting for the effects of dip, the mean vectors from all the rocks of a similar age were combined by taking the mean of the direction cosines, and a statistical estimate of the dispersion was made by Fisher's method. These group mean positions were then used to give the position of the magnetic pole[3]. The main computations were carried out on the Cambridge digital computor *Edsac*.

Table 1

Group	No. of rocks	Dip corrected: D	I	α	κ	Position of pole Lat.	Long.	$d\chi$	$d\psi$
Middle and Upper Trias	44	358	68	3	41	160E	68N	5	4
Permian	10	175	−37	7	50	163E	38N	8	5
Tillite Formation	16	135	16	18	5	161W	04N	19	10
Multicoloured Series	10	110	36	9	31	141W	13S	10	5

Explanation of columns:
D, declination in degrees measured from true north.
I, inclination to the horizontal in degrees, positive downwards, negative upwards.
α, semi-angle of the cone of confidence drawn at 95 per cent probability-level.
κ, estimate of precision.
$d\chi$, $d\psi$, semi-axes of the ovals of confidence drawn at the 95 per cent probability-level.

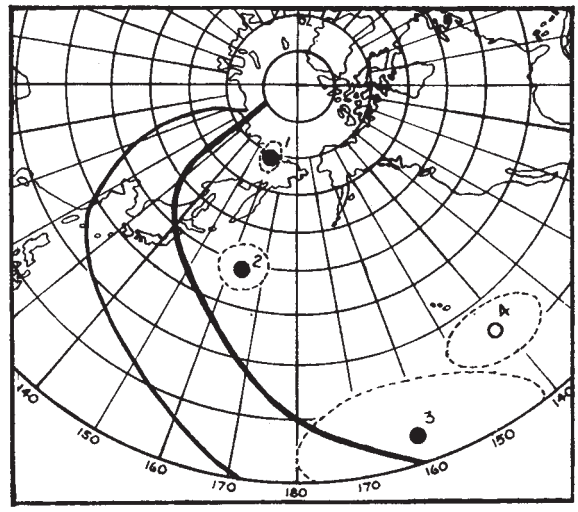


Fig. 1. Polar projection showing positions of magnetic poles derived from (1) Middle and Upper Trias; (2) Permian; (3) Tillite Formation; (4) Multicoloured series. ●, pole plotting in northern hemisphere; ○, pole plotting in southern hemisphere; broken lines show ovals of confidence drawn at the 95 per cent level. For comparison, the approximate path of north pole inferred from British rocks (thick line) and the path inferred from American rocks (thin line) are shown, after Creer *et al.* (ref. 5)

Table 1 lists the group mean positions and pole positions; the latter are shown graphically in Fig. 1.

(1) The Middle and Upper Trias rocks were sampled at Kapp Biot, where 44 specimens of flat-lying red sandstones and marlstones were collected. These were found to be strongly and uniformly magnetized, with eight of the rocks forming a reversed group directed upwards and to the south, whereas the remainder formed a normal group downwards and to the north. The two groups have been combined to give the group mean position.

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(2) Ten flat-lying red sandstones of Permian age collected from the two well-separated areas of Gauss Halvö and Mesters Vig were strongly and uniformly magnetized to the south and upwards, forming a reversed group at a shallower angle than the Middle and Upper Trias.

(3) The tillite formation of uppermost Pre-Cambrian age was sampled on both limbs of an anticline on Ella Ø; many of the fifty-four rocks collected proved too weakly magnetized to be measured, or had vectors close to the present field. Sixteen of these rocks have been combined to give the group mean, on the grounds that they show positive evidence of stability of magnetization by the presence of reversals, low scattering of vector positions between rocks from the same collecting station, and by the use of the tectonic check provided by the anticline. The rocks are of varied lithology, ranging from red sandstones and tillites to grey and brown shales.

(4) The multicoloured series (3,500 m. below basal Cambrian) form part of the Eleonore Bay Formation, which is a thick conformable sequence passing upwards through the tillite formation into rocks of Cambrian age. Ten specimens of red mudstone were collected from two outcrops on the south side of Ruth Ø, where the beds dip gently south-east. The vectors fall into a single well-defined group directed downward to the east.

The polar stereographic projection (Fig. 1) shows the positions of the pole computed from the four sets of vectors. These pole positions fall into the same general area of the Pacific Ocean and Asia as the pole positions obtained by other workers for rocks collected in North America and Europe, and the pole positions of the older rocks show the same tendency to approach the equator, apparently crossing it at a time between the tillite formation and the Upper Eleonore Bay Formation. This is in agreement with results for rocks of similar age from Europe[3], and we[4] have already discussed the implications of this equatorial position coinciding with a major glaciation. The pole positions for these Greenland rocks lie generally about 18° east of the general European trend, or about 30° east of the North American trend, which would be consistent with a post-Trias movement of Greenland away from North America.

Taking into account this postulated movement between the continents and the similar movement that has been suggested between Europe and North America[5], the pole positions derived from rocks of similar age from the three areas can be compared. In the post-Cambrian rocks there is some agreement, and if the comparison is extended into the Pre-Cambrian, the pole position derived from the Greenland Upper Eleonore Bay rocks lies intermediate between the pole positions from the Cambrian and Upper Torridonian of Europe, and between the Cambrian and the late Pre-Cambrian positions from the Upper Keweenawan and Beltian of North America, which accords with the stratigraphic relationship of the Upper Eleonore Bay Series to the overlying Cambrian rocks.

We wish to correct an error in quoting E. Irving's work. The second sentence in our recent communication[4] should read "Palæomagnetic studies in Australia suggest that Permian Tillites were near the . . . pole".

[1] Harland, W. B., *Polar Record*, **9**, 558 (1959).
[2] Bidgood, D. E. T., and Harland, W. B., *Bull. Geol. Soc. Amer.*, **70**, 641 (1959).
[3] Creer, K. M., *Phil. Trans. Roy. Soc.*, A, **250**, 123 (1957).
[4] Harland, W. B., and Bidgood, D. E. T., *Nature*, **184**, 1860 (1959).
[5] Creer, K. M., *et al.*, *Phil. Trans. Roy. Soc.*, A, **250**, 144 (1957).

USE OF RADIOACTIVE NITROGEN IN STUDYING NITROGEN FIXATION IN BACTERIAL CELLS AND THEIR EXTRACTS

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THE stable isotope of nitrogen (nitrogen-15) has been successfully employed to follow nitrogen fixation in soil bacteria, blue-green algae and in root nodules of legumes and non-legumes[1,2]. It has also been used recently to demonstrate the incorporation of nitrogen into cell-free extracts of *Clostridium pasteurianum*[3], *Azotobacter vinelandii*[4,5], blue-green algae[6] and those of *Chromatium*[7]. Since the nitrogen-15 technique is time-consuming, some two days being required to complete an experiment from the initial exposure to the gas to the final analysis in a mass spectrometer, we have explored the possibility of using radioactive nitrogen (nitrogen-13, β^+ 1·24 MeV.; half-life 10·05 min.) to study nitrogen fixation in bacterial cells and cell-free extracts prepared from them. We present here details of the technique used together with some of the preliminary results.

Nitrogen-13 has been used previously in chemical[8,9] and biological studies[10]. Carangal and Varner[11], after bombarding thin graphite targets with protons or deuterons in a cyclotron, digested them in acid mixtures and prepared labelled nitrate, nitrite and ammonium compounds which were then supplied to excised plant tissues. Although these workers produced molecular nitrogen-13 by oxidizing labelled ammonia with sodium hypobromite, they did not report its use in biological nitrogen fixation.

In our experiments, nitrogen-13 was produced continuously in the Medical Reseach Council's cyclotron by bombarding coarsely granulated charcoal, retained behind a stainless steel mesh, with

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